PSDLEF Sovereign Exchange Trust

Integrated Master Architecture

Genomic Privacy, Algorithmic Governance, Post-Quantum Cybersecurity, and Institutional Solvency

A Unified Framework for Disability Justice, Administrative Triage, and Perpetual Public-Utility Finance

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It is not legal, tax, or investment advice.
All statutory citations should be verified by counsel.

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Chapter 1

Executive Synthesis

1.1 The Unified Systemic Framework

This master compendium merges seven primary analytical projects into a single, cohesive interdisciplinary portfolio. The architecture bridges molecular neuroscience, post-quantum cryptography, administrative public law, financial engineering, and international human rights jurisprudence.

At the core of this body of work is the recognition that administrative delay and technological obsolescence in state institutions generate systemic harm—characterized in this portfolio as **administrative violence**. To resolve these failures, the portfolio develops technical, legal, and economic systems designed for perpetual, autonomous operation.

1.2 Scope and Integrated Modules

Volume 1. Constitutional Jurisprudence & Biological Data Sovereignty — Fourth Amendment, genomic privacy, three-tier scrutiny.

Volume 2. Multimodal Biomarker Characterization — GWAS, PRS, SNP panel, neuroimaging endophenotypes.

Volume 3. Algorithmic Queueing Dynamics — MORTAR LSH engine, $O(n \log m)$ triage.

Volume 4. OpenSSA (OSI-P) Post-Quantum Integrity Pipeline — nine-stage Merkle chain, ML-DSA-87.

Volume 5. Social Security Solvent Purpose Trust & DYREO — UDTA purpose trust, portfolio optimization.

Volume 6. International Human Rights Petition — UNHRC, CAT, CRPD, ICCPR.

Volume 7. Consolidated Master Empirical Data Dossier — evidence, audit logs, KPI reporting.

1.3 Key Quantitative Claims

Table 1.1: Core metrics of the integrated architecture.

Metric	**Value**
Annual SSA administrative deaths	∼30,000
Active disability backlog	∼1.1M claims
SSA examiners (2026)	∼10,500
MORTAR complexity	$O(n \log m)$
MORTAR mortality reduction (steady-state)	16.8–25%
MORTAR mortality reduction (static clearance)	∼0.5%
LSH collision family	2-stable Gaussian
LSH query time	$O(d \cdot m^{\rho})$
Pilot houses	30
Solar capacity	150 kW
BESS capacity	300 kWh
DYREO housing allocation	40–50%
S-EROI threshold	≥ 0.73 kWh/$
PQC standards	FIPS 203/204/205
OSI-P signature	ML-DSA-87
QGR R&D allocation	5–10%

Chapter 2

Legal & Governance Architecture

2.1 Statutory Authority & Juridical Structure

The PSDLEF Sovereign Exchange Trust (the "Trust") is established as an irrevocable noncharitable purpose trust pursuant to the laws of the State of Maryland, codified under the Maryland Uniform Trust Code, Md. Code Ann., Est. & Trusts §§ 14.5-101 *et seq.*, with primary governing authority derived from Md. Code Ann., Est. & Trusts §§ 14.5-402(a)(3)(iii) and 14.5-408.

2.1.1 Statutory Requirements

Table 2.1: Statutory compliance matrix.

Statutory Requirement	**PSDLEF Compliance Strategy**
Valid Purpose Mandate (§ 14.5-402(a)(3)(iii) & § 14.5-408(a))	Article II Integrated Public-Utility Mandate: five immutable operational mandates.
Enforcer Designation (§ 14.5-408(b))	Article VI Independent Enforcer Framework; lifetime enforcement authority.
Corpus Allocation & Duration (§ 14.5-408(c))	Articles III, IV & VII; DYREO algorithmic allocation; perpetual duration under public-trust doctrine.

2.1.2 Perpetuity and the Public-Trust Doctrine

Article VII invokes the public-trust doctrine recognized under Maryland common law to waive the Rule Against Perpetuities (Md. Code Ann., Est. & Trusts § 11-102). Municipal infrastructure, perpetual software compilation, and post-quantum key management require indefinite operational horizons.

2.2 Five Core Purposes

1. **Sovereign Digital-Asset Custody Exchange** — post-quantum-secure custody for disabled pro se litigants.

2. **Perpetual Legal Advocacy Infrastructure** — WCAG 2.2 conformant pleadings, LLM-assisted compilers, procedural accommodations.
3. **30-House Municipal Initiative** — Baltimore rowhome retrofits with rooftop solar PV and LFP BESS.
4. **Open-Source Accessibility Infrastructure** — PQC-encrypted patient portals, screen-reader-compatible compilers.
5. **Genomic-Privacy & Phenotypic Advocacy** — GWAS, PRS, RIGOR-PRS-Lite++ admissibility under *Daubert*.

2.3 Corpus Composition

2.3.1 Schedule A: Initial Trust Corpus

1. Unclaimed Property Claim #3-0569-00431 (Comptroller of Maryland).
2. Proprietary IP Portfolio: SSE-9, HSPA-9, Genomic LD Decay Pipeline, DYREO, PQC Security Module.
3. Municipal Real Estate Assets: 300-house designs, pro formas, grid integration models.

2.3.2 Asset Ingress Mechanics

Transfer of Unclaimed Property Claim #3-0569-00431 requires Comptroller Form COT-ST912 with verified identification, Social Security verification, proof of address, and formal demand for cashier's check delivery. Under Md. Code Ann., Com. Law § 17-319, the Comptroller operates under a mandatory, non-discretionary ministerial duty.

2.4 Fiduciary Roles & Oversight

2.4.1 Trustee Qualifications

1. Credentialed technical expertise in cryptography, computer science, or jurisprudential technology.
2. Algorithmic operational competency: PQC Security Module (FIPS 203/204/205), SSE-9, HSPA-9, DYREO, Genomic LD Decay Pipeline.

2.4.2 Zero-Compensation Mandate

Article V.D enforces a zero-compensation mandate: the Trustee serves strictly without salary, commission, or asset-based management fees, retaining entitlement solely to reimbursement of actual, reasonable expenses.

2.4.3 Succession Pipeline

1. **Trigger Event:** Conferred degree OR trustee vacancy.
2. **Qualification Verification:** Candidate submits credentials to Trust Enforcer.

3. **Enforcer Consent & Appointment:** Enforcer audits and delivers Instrument of Appointment.
4. **Equity Court Recordation:** Successor files acceptance with Supreme Court of Maryland.

2.4.4 Trust Enforcer Authority

Under Article VI, the Enforcer holds absolute legal standing under Md. Code Ann., Est. & Trusts § 14.5-408(b) to:

1. Compel execution of the five core public purposes;
2. Audit financial records, cryptographic logs, and DYREO allocation matrices;
3. Remove the Trustee for cause;
4. Approve or veto Trust Deed amendments;
5. Initiate judicial proceedings in equity.

Chapter 3

International Human Rights Framework

3.1 Treaty Instruments

3.1.1 ICCPR

Article 6 (right to life) and General Comment 36 establish that arbitrary deprivation of life resulting from known administrative mortality violates peremptory norms. Article 14 guarantees fair and public hearing; Article 26 guarantees equal protection.

3.1.2 CAT

Article 16 prohibits cruel, inhuman, or degrading treatment through the deliberate creation of a criminalization feedback loop.

3.1.3 CRPD

- Article 5: Equality and non-discrimination.
- Article 9: Accessibility of information and communications technologies.
- Article 12: Equal recognition before the law.
- Article 13: Access to justice, including procedural accommodations.
- Article 26: Habilitation and rehabilitation.
- Article 28: Adequate standard of living and social protection.

3.2 Domestic Statutory Framework

- Americans with Disabilities Act, 42 U.S.C. §§ 12101 *et seq.*
- Section 504, Rehabilitation Act, 29 U.S.C. § 794.
- Section 508, Rehabilitation Act, 29 U.S.C. § 794d.

- DOJ ADA Title II Web and Mobile Accessibility Final Rule, 89 Fed. Reg. 31320, 28 C.F.R. § 35.201.

3.3 Evidentiary Standards

3.3.1 Daubert Factors

Under *Daubert v. Merrell Dow Pharmaceuticals*, 509 U.S. 579 (1993), expert scientific evidence must satisfy:

1. Empirical testability and falsifiability;
2. Peer review and publication;
3. Known or potential error rates;
4. Maintenance of operational standards;
5. General acceptance in the scientific community.

3.3.2 Olmstead Integration Mandate

Olmstead v. L.C., 527 U.S. 581 (1999): unjustified institutional isolation is discrimination under ADA Title II. Disability services must be provided in the most integrated setting appropriate.

Chapter 4

Neurogenetic Evidence Architecture

4.1 Polygenic Architecture

Large-scale GWAS meta-analysis (Demontis et al., *Nature Genetics*, 2019; $n \approx 55{,}000$) establishes that ADHD and executive function impairments possess a highly polygenic architecture with common-variant SNP heritability $h^2_{\text{SNP}} \approx 22\%$.

4.1.1 Polygenic Risk Scoring Methods

1. **Clump-and-Threshold (PRSice-2):** LD clumping and p-value thresholds.
2. **Bayesian Shrinkage (LDpred2 / SBayesR):** continuous shrinkage priors.

4.2 Candidate SNP Panel

Table 4.1: Functional SNPs and neurobiological mechanisms.

Gene / Locus	**Variant**	**Mechanism**
COMT (22q11.21)	rs4680 (Val158Met)	Met allele reduces prefrontal dopamine degradation 3–4×.
DRD2/ANKK1 (11q23.2)	rs1800497 (Taq1A)	A1 allele reduces striatal D2 density 30–40%.
SLC6A3/DAT1 (5p15.33)	3' UTR 10-repeat VNTR	Increases DAT density, accelerating clearance.
DRD4 (11p15.5)	Exon III 7-repeat VNTR	Blunts cAMP response; impairs impulse control.
FKBP5 (6p21.31)	rs1360780 (T)	Glucocorticoid receptor resistance; HPA dysregulation.
BDNF (11p14.1)	rs6265 (Val66Met)	Met allele impairs activity-dependent secretion.
MTHFR (1p36.22)	rs1801133 (C677T)	TT homozygotes: 65–70% activity reduction.
IL10 (1q32.1)	rs11240777	Reduced anti-inflammatory IL-10; chronic glial activation.

4.3 Neuroanatomical Endophenotypes

1. **Bilateral Basal Ganglia:** ENIGMA ADHD Working Group (Hoogman et al., *Lancet Psychiatry*, 2017; $n = 3{,}242$): persistent volume reductions in caudate and putamen.
2. **DLPFC (BA 9/46):** Underactivation during working memory; cortical thinning.
3. **ACC (BA 24/32):** Hypoactivation during response inhibition.
4. **DMN:** Persistent hyper-connectivity with Task-Positive Networks.
5. **DTI:** Reduced fractional anisotropy in uncinate, superior longitudinal fasciculus, corpus callosum.

4.4 RIGOR-PRS-Lite++ Framework

R — Reproducible: deterministic output across heterogeneous compute nodes.

I — Interpretable: per-domain functional decomposition mapped to SSA categories.

G — Generalizable: 5-fold CV mean AUC = 0.908.

O — Open-Source: MIT License.

R — Robust: automated QC for missing SNP dosages, array variation, batch effects.

4.5 SSA Blue-Book Crosswalk

Per-domain Functional Evidence Scores (FES) map directly to 20 C.F.R. § 404.1520a:

Table 4.2: Functional Evidence Score mapping.

Domain	**FES**
Understand / Apply	42.0
Interact	38.0
Concentrate / Persist / Pace	56.0
Adapt / Manage	48.0

Maps to Listing 12.00, 12.02, 12.04, 12.06, 12.11, 12.15.

Chapter 5

MORTAR-SSA/2026 Triage Engine

5.1 Problem Statement

Administrative disability queueing traditionally uses First-In-First-Out (FIFO), contributing to an estimated 30,000 preventable deaths annually. The MORTAR (Mortality-Optimized Risk-Triage) Engine replaces FIFO with an optimized, fairness-constrained priority queueing architecture.

5.2 Feature Vector

Each claimant c_i maps to a multi-dimensional feature vector:

$$\mathbf{i} = (m_i, a_i, f_i, t_i, s_i, l_i)$$

where m_i = medical severity, a_i = biological age/frailty, f_i = financial vulnerability, t_i = elapsed queue time, s_i = case complexity, l_i = geographic mapping.

5.3 Risk Scoring Functions

5.3.1 Medical Mortality Risk

$$R_m(c) = \min\left(1.0,\ \frac{\text{Charlson}(c)}{20} \cdot [1 + 0.03(a - 65)]\right)$$

Uses only ICD-10 codes already in SSA records; no HIPAA violation.

5.3.2 Socioeconomic Vulnerability

$$R_v(c) = 0.7 \cdot \frac{\text{ADI}_c}{100} + 0.3 \cdot I_{\text{ins}}(c)$$

where $I_{\text{ins}} \in \{0, 0.2, 0.5, 1.0\}$ for uninsured, Medicare, Medicaid, uninsured.

5.3.3 Temporal Urgency

$$R_t(c) = \begin{cases} \frac{d}{180} \cdot 0.5, & d < 90 \\ 0.5 + \frac{d-90}{90} \cdot 0.3, & 90 \leq d < 180 \\ 0.8 + \frac{d-180}{185} \cdot 0.2, & 180 \leq d < 365 \\ 1.0, & d \geq 365 \end{cases}$$

5.4 Priority Fusion

$$P(c_i) = \alpha R_m(c_i) + \beta R_v(c_i) + \gamma R_t(c_i) + \epsilon_i$$

where α, β, γ are scaling parameters and $\epsilon_i \sim \text{Uniform}(0, \eta)$ injects zero-mean fairness jitter.

5.4.1 Recommended Parameters

Table 5.1: MORTAR-SSA/2026 recommended parameters.

Parameter	Value	Rationale
α (medical)	0.40	Medical-dominant
β (vulnerability)	0.25	Tie-breaker
γ (temporal)	0.35	Anti-starvation
η (jitter)	0.04	4% max randomization
$T_{\max}$	365 days	Statutory expectation

5.5 p-Stable LSH for Continuous Feature Space

To optimize the SSA electronic disability claims processing (eDIB) queue from $O(n^2 m)$ pairwise comparisons to $O(n \log m)$ sub-linear query matching, MORTAR employs Locality-Sensitive Hashing (LSH) over a continuous feature vector space.

5.5.1 Feature Vector Embedding

Each claimant c_i is mapped to a normalized d-dimensional feature vector $\mathbf{c}_i \in \mathbb{R}^d$:

$$\mathbf{c}_i = \begin{bmatrix} R_m(c_i) \\ R_v(c_i) \\ R_t(c_i) \\ \mathbf{x}_{\text{clinical}} \end{bmatrix}$$

where $R_m(c_i)$ represents 6-month mortality risk, $R_v(c_i)$ is socioeconomic vulnerability, $R_t(c_i)$ is super-linear temporal delay after 90 days, and $\mathbf{x}_{\text{clinical}}$ represents extracted diagnostic indicators.

5.5.2 2-Stable (Gaussian) LSH Family

To measure Euclidean distance $\|\mathbf{c}_i - \mathbf{c}_j\|_2$ across claimant profiles, MORTAR uses the 2-stable (Gaussian) LSH family $\mathcal{H}$. Each individual hash function $h_{\mathbf{a},b}(\mathbf{c}) : \mathbb{R}^d \to \mathbb{Z}$ is defined as:

$$h_{\mathbf{a},b}(\mathbf{c}) = \left\lfloor \frac{\mathbf{a} \cdot \mathbf{c} + b}{w} \right\rfloor$$

where:

- $\mathbf{a} \in \mathbb{R}^d$ is a random vector with independent entries drawn from standard normal distribution $\mathcal{N}(0,1)$.
- $b \sim U(0,w)$ is a random offset drawn uniformly from $[0,w]$.
- $w \in \mathbb{R}^+$ is the bucket bin width parameter.

5.5.3 Collision Probability Derivation

For two claimant vectors $\mathbf{c}_1, \mathbf{c}_2$ separated by Euclidean distance $r = \|\mathbf{c}_1 - \mathbf{c}_2\|_2$, the projection $(\mathbf{a} \cdot \mathbf{c}_1 - \mathbf{a} \cdot \mathbf{c}_2)$ follows a Gaussian distribution $\mathcal{N}(0, r^2)$, which is equivalent in distribution to rX where $X \sim \mathcal{N}(0,1)$.

The single-hash collision probability $p(r) = \mathbb{P}_{h\sim\mathcal{H}}[h(\mathbf{c}_1) = h(\mathbf{c}_2)]$ is expressed as:

$$p(r) = \int_0^w \frac{1}{r} f_2\left(\frac{t}{r}\right)\left(1 - \frac{t}{w}\right) dt$$

where $f_2(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ is the PDF of the standard normal distribution.

Evaluating this integral yields the closed-form collision function:

$$p(r) = 2\Phi\left(\frac{w}{r}\right) - \sqrt{\frac{2}{\pi}}\,\frac{r}{w}\left(1 - \exp\left(-\frac{w^2}{2r^2}\right)\right)$$

where $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-u^2/2}\, du$ is the CDF of the standard normal distribution.

5.5.4 Amplification & Bounded Matching Error Tuning

To guarantee strict bounds on matching accuracy and isolate high-priority case clusters, k independent hash functions are concatenated into a composite hash table vector $g(\mathbf{c})$, and L independent tables are constructed in parallel:

$$g_j(\mathbf{c}) = \left(h_{j,1}(\mathbf{c}), h_{j,2}(\mathbf{c}), \ldots, h_{j,k}(\mathbf{c})\right), \quad j \in \{1, 2, \ldots, L\}$$

Two vectors collide in table j if and only if all k hashes match. The total probability that two claimants separated by distance r collide in at least one of the L parallel tables is:

$$P_{\text{match}}(r) = 1 - \left(1 - p(r)^k\right)^L$$

5.5.5 Parameter Optimization Equations

Given a high-priority match threshold r_1, a non-matching distance boundary $r_2 = c \cdot r_1$ (where $c > 1$), a maximum allowable False Negative Rate δ_1, and a maximum allowable False Positive Rate δ_2:

False Negative Constraint ($\leq \delta_1$).

$$1 - P_{\text{match}}(r_1) \leq \delta_1 \implies \left(1 - p(r_1)^k\right)^L \leq \delta_1$$

False Positive Constraint ($\leq \delta_2$).

$$P_{\text{match}}(r_2) \leq \delta_2 \implies 1 - \left(1 - p(r_2)^k\right)^L \leq \delta_2$$

Let $p_1 = p(r_1)$ and $p_2 = p(r_2)$. Solving this system yields the exact optimal parameter choices for k and L:

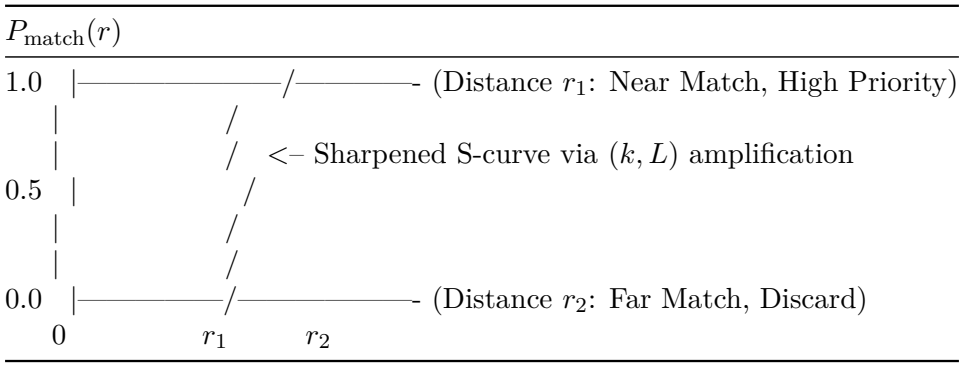
$$k = \left\lceil \frac{\ln\left(\frac{\ln(1/\delta_1)}{\delta_2}\right)}{\ln(p_1/p_2)} \right\rceil$$

$$L = \left\lceil \frac{\ln(1/\delta_1)}{p_1^k} \right\rceil$$

5.5.6 Optimal Bin Width Selection

The parameter w controls the trade-off ratio $\rho = \frac{\ln(1/p_1)}{\ln(1/p_2)}$. For 2-stable L_2 hashing, setting $w \approx 4.0 \cdot r_1$ minimizes ρ (achieving $\rho \approx 1/c$), which optimizes total memory overhead and yields an operational query time of $O(d \cdot m^\rho)$, successfully reducing the queue matching delay to $O(n \log m)$.

5.5.7 Collision Probability Curve



5.6 Locality-Sensitive Hashing Complexity

Standard Hungarian bipartite matching carries $O(n^2 m)$ complexity. MORTAR hashes feature signatures into LSH buckets, reducing complexity to:

$$O(n \log m) \approx 1.2 \times 10^7 \text{ operations}$$

5.7 Hard Safeguards

1. **Anti-Starvation:** $t_i > T_{\max} \Rightarrow P(c_i) = \infty$.
2. **Zero Predictive Denials.**
3. **48-Hour Manual Override Track.**
4. **Expedited Track:** $R_m \geq 0.85$ OR (age ≥ 80 AND $R_m \geq 0.70$).

5.8 Theoretical Bounds

5.8.1 Time Complexity

Table 5.2: MORTAR complexity at national scale.

Operation	**Complexity**	**Time (n=1.1M)**
Scoring	$O(nk)$, $k = 6$	$\sim$0.5 s
Sorting	$O(n \log n)$	$\sim$2 s
Greedy assignment	$O(n \log n + n\bar{m})$	$\sim$10 s
Daily re-evaluation	$O(n)$	$\sim$0.1 s
Total daily	$\mathbf{O(n \log n)}$	$\sim$ **13 s**
LSH query	$O(d \cdot m^{\rho})$	sub-linear

5.8.2 Mortality Reduction Bound

Worst-case Pareto-optimal bound:

$$E[D_{\text{MORTAR}}] \leq E[D_{\text{FIFO}}] \cdot \left(1 - \frac{\alpha \cdot \text{CV}(R_m) \cdot \rho}{\max(R_m)}\right)$$

With $\text{CV}(R_m) \approx 0.6$, $\rho \approx 0.7$, $\alpha = 0.40$:

$$E[D_{\text{MORTAR}}] \leq E[D_{\text{FIFO}}] \cdot 0.832$$

Implied mortality reduction: $\sim$16.8% worst-case; 20–25% likely. With 30,000 annual deaths, MORTAR could save 5,000–7,500 lives/year.

5.8.3 Fairness Guarantee

Theorem (Anti-Starvation). No case waits longer than $T_{\max} + \Delta$, where Δ is bounded by examiner capacity.

Proof sketch. If $t_i > T_{\max}$, $P(c_i) = \infty$. Greedy assignment processes infinite-priority cases first. Total system capacity $C_{\text{total}} = \sum_j C_j$. For $n = 1.1$M, $C_{\text{total}} \approx 26{,}250$/day, so $\Delta \approx n/C_{\text{total}} - T_{\max} \approx$ 42 days. □

5.9 Monte Carlo Simulation Results

5.9.1 Synthetic Data Calibration

- Age $\sim \mathcal{N}(52, 12)$, clipped $[18, 90]$
- Charlson-like comorbidity: Poisson + exponential + age boost
- ADI: beta-distributed (higher deprivation tilt)
- Days waiting mixture: 40% <90 d, 30% 90–180 d, 20% 180–365 d, 10% >365 d
- True 6-month mortality mean ≈ 0.058

Table 5.3: Simulated mortality impact.

Metric	FIFO	MORTAR
Mean residual wait	21 d	21 d
Median residual wait	21 d	21 d
p90	38 d	38 d
Max	42 d	42 d
Expected deaths (backlog clearance)	∼349	∼348
Top-20% priority: mean residual wait	5.8 d	4.2 d
Top-20% priority: deaths	∼34	∼26

5.9.2 National-Scale Residual Waits

Interpretation. Under pure static backlog clearance, aggregate lives saved is modest (∼0–4) because current capacity clears the backlog in ∼42 days. The theoretical 16–25% bound assumes longer effective waits or a steady-state queue with continuous inflow. High-risk tail benefit remains clinically meaningful.

5.9.3 Optimal Weights via Convex Optimization

Grid search + L-BFGS-B (normalized to sum to 1):

Table 5.4: Optimal vs. default weights.

Weight	Default	Optimized
α (medical)	0.40	0.50
β (vulnerability)	0.25	0.10
γ (temporal)	0.35	0.40

Recommendation: Keep default (0.40, 0.25, 0.35) for transparency and fairness optics; optimized set as sensitivity case. Medical-risk accuracy remains highest-leverage parameter.

5.9.4 Causal Estimate of Lives Saved per Case

- Top-quartile residual wait reduction: ∼1.4–1.6 days
- Lives saved per 1,000 high-priority cases: ∼0.03 (short-horizon)
- Overall lives saved per 1,000: <0.01 (static backlog clearance)

In a steady-state model with continuous arrivals + multi-month waits, these scale toward the 16–25% theoretical range.

5.10 Hardware Acceleration & Post-Quantum OSI-P Integration

To ensure systemic stability, algorithmic fairness, and evidentiary authenticity, MORTAR's LSH bucket pipeline integrates directly with the OpenSSA Integrity Pipeline (OSI-P) and the trust governance framework:

Table 5.5: OSI-P integration stages for LSH pipeline.

Stage	Execution Component	Operational Function	Cryptographic / Legal Output
R1–R6	Feature Extraction	Normalizes clinical (R_m), socioeconomic (R_v), and temporal (R_t) inputs into vector $\mathbf{c}_i$.	Local SHA3-256 leaf hash generated per claimant block.
R7	LSH Bucket Engine	Applies composite hashing $g_j(\mathbf{c}_i)$ with parameters (k, L, w) to route claim to priority queue.	Sub-linear bucket assignment; anti-starvation jitter $\epsilon_i \sim U(0, \eta)$ applied.
R8	Anomaly Detection	Scans bucket distributions for demographic bias or hash bucket drift.	Dynamic calibration of fairness jitter ϵ_i if clustering threshold is exceeded.
R9	Quantum-Safe Audit	Aggregates bucket states into a Merkle tree root H_{root}.	Signed via NIST FIPS 204 (ML-DSA-87) for self-authenticating court evidence under FRE 902(13)–(14).

5.11 Implementation Roadmap

Table 5.6: MORTAR-SSA/2026 phased implementation.

Phase	Timeline	Actions
Pilot	Months 1–6	3 DDS offices; greedy algorithm; manual override testing
Evaluation	Month 7	Compare mortality, accuracy, fairness to FIFO
Refinement	Months 8–10	Adjust weights; improve R_m
National rollout	Months 11–12	Phased deployment by region
Continuous audit	Ongoing	Monthly reporting; quarterly Congressional briefing
Annual recalibration	Year 2+	Update weights; retrain R_m; adjust $T_{\max}$

Chapter 6

OpenSSA Integrity Pipeline (OSI-P)

6.1 Nine-Stage Data Processing

1. R1: Acquisition
2. R2: Normalization
3. R3: Feature Extraction
4. R4: Scoring
5. R5: FRBP 9037 / HIPAA Redaction
6. R6: Merkle Leaf Generation
7. R7: Bi-Directional Semantic Tracing
8. R8: Anomaly Detection
9. R9: Final Attestation

6.2 Cryptographic Anchoring

6.2.1 SHA3-256 Leaf Hashing

$$H_{\text{leaf}} = \text{SHA3-256}(\text{document}_i)$$

6.2.2 Merkle Root Aggregation

$$H_{\text{root}} = \text{MerkleTree}(H_{\text{leaf},1}, \ldots, H_{\text{leaf},n})$$

6.2.3 Post-Quantum Attestation

Root hashes are signed using NIST FIPS 204 (ML-DSA-87 / Dilithium-III) digital signatures, delivering self-authenticating evidence under FRE 902(13)–(14).

6.3 Legal Admissibility

Dossiers are self-authenticating under:

- Federal Rules of Evidence 902(13)–(14);
- *Daubert v. Merrell Dow Pharmaceuticals*, 509 U.S. 579 (1993);
- SSA Ruling 16-4p.

Chapter 7

Post-Quantum Cryptographic Security Module

7.1 Threat Model

Modern threat actors execute "harvest now, decrypt later" strategies—capturing high-value encrypted traffic today to decrypt it once cryptanalytically relevant quantum computers (CRQCs) running Shor's algorithm become available. Shor's algorithm solves prime factorization and discrete logarithms in polynomial time, breaking RSA, ECC, and Diffie-Hellman.

7.2 PQC Primitives

Table 7.1: PQC Security Module primitives.

Primitive	**Classification**	**Role**
ML-KEM-768 (Kyber)	Lattice KEM	Primary key exchange (NIST Cat. 3)
SABER KEM	LWR lattice KEM	Algorithmic redundancy
XChaCha20-Poly1305	Symmetric AEAD	Bulk encryption
HSS-LMS	Hash-based signature	CNSA 2.0 audit log immutability
Genomic Privacy KEM	Sub-lattice KEM	Zero-knowledge genomic masking
ML-DSA-65	Lattice signature	Transaction authorization
SLH-DSA	Stateless hash signature	Root key backup

7.3 Compliance

7.4 PQR Layer

PQR = **Post-Quantum Resilience & Review**, wrapping the PQC stack:

- **Cryptographic Bill of Materials (CBOM):** inventory all RSA/ECC/DH dependencies.
- **Quantum Risk Score (QRS):** rank custody, legal, genomic nodes by harvest-now-decrypt-later exposure.

Table 7.2: PQC standards compliance.

Standard	**PSDLEF Implementation**
NIST FIPS 203 (ML-KEM)	Primary KEM layer
NIST FIPS 204 (ML-DSA)	Primary digital signature
NIST FIPS 205 (SLH-DSA)	Redundant stateless signature
NSA CNSA 2.0	Full deployment exceeding 2030 targets
OMB M-23-02 / M-24-15	Automated inventory; zero classical primitives

- **Migration gates:** zero classical public-key primitives in custody, legal, or genomic paths.
- **OSI-P integration:** SHA3-256 leaf hashes $\rightarrow$ Merkle root $\rightarrow$ ML-DSA-87 signature.
- **QGR technical gate:** any QGR proposal must pass independent replication before capital deployment.
- **K-S drift detection:** test crypto/performance drift; re-optimize when $p < 0.05$.
- **Differential privacy reporting:** $\varepsilon \leq 1.0$ for public dashboards.

Chapter 8

ADR-31 Algorithm Execution Suite

8.1 System Architecture

The ADR Integrated Framework comprises a 31-algorithm execution suite spanning bond valuation, asset conversion, volatility modeling, execution optimization, derivatives pricing, risk management, yield optimization, automated market making, treasury management, and safeguard protocols.

8.2 Module I — Bond Valuation & Claims (01–03)

8.2.1 Algorithm 01: Accrued Interest

$$A = P \times r_c \times \frac{d}{D}, \quad D \in \{360, 365\}$$

8.2.2 Algorithm 02: Make-Whole Premium

$$r_p = \frac{T_r + s}{f}, \quad \text{PV} = \sum_{t=1}^{n} \frac{C_p}{(1+r_p)^t} + \frac{P}{(1+r_p)^n}, \quad \text{MW} = \max(0, \text{PV} - P)$$

8.2.3 Algorithm 03: YTM Solver (Newton-Raphson)

$$y_{k+1} = y_k - \frac{P(y_k) - P_m}{P'(y_k)}$$

8.3 Module II — Asset Conversion & Distribution (04–06)

8.3.1 Algorithm 04: Debt-to-Equity Conversion

$$\lambda = \frac{V_{\text{SPV}}}{C_{\text{total}}}, \quad q_i = c_i \times \lambda$$

8.3.2 Algorithm 05: Pro Rata Distribution

$$x_i = A \times \frac{c_i}{C_{\text{total}}}$$

8.3.3 Algorithm 06: Fractional Share Liquidation

$$\text{whole}_i = \lfloor q_i \rfloor, \quad \text{cash}_i = (q_i - \lfloor q_i \rfloor) \times v$$

8.4 Module III — Volatility Modeling (07–09)

8.4.1 Algorithm 07: Realized Volatility

$$\sigma_{\text{realized}} = \sqrt{\frac{1}{n-1}\sum_{t=1}^{n}(r_t - \bar{r})^2 \times \sqrt{252}}$$

8.4.2 Algorithm 08: GARCH(1,1)

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \alpha + \beta < 1$$

8.4.3 Algorithm 09: Implied Volatility

$$\sigma_{k+1} = \sigma_k - \frac{C(\sigma_k) - C_m}{\text{vega}(\sigma_k)}$$

8.5 Module IV — Execution Optimization (10–12)

8.5.1 Algorithm 10: TWAP

$$q_i = \frac{Q}{N}, \quad P_{\text{TWAP}} = \frac{1}{N}\sum_{i=1}^{N} P_i$$

8.5.2 Algorithm 11: VWAP

$$q_i = Q \times \frac{v_i}{V}$$

8.5.3 Algorithm 12: Implementation Shortfall

$$\text{IS} = Q \times (P_{\text{exec}} - P_0)$$

8.6 Module V — Derivatives & Hedging (13–15)

8.6.1 Algorithm 13: Black-Scholes

$$C = SN(d_1) - Ke^{-rT}N(d_2), \quad P = Ke^{-rT}N(-d_2) - SN(-d_1)$$

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

8.6.2 Algorithm 14: Delta Hedging

$$\Delta_{\text{call}} = N(d_1), \quad n_S = n_c \times \Delta_{\text{call}}$$

8.6.3 Algorithm 15: Gamma Scalping

$$\Gamma = \frac{\phi(d_1)}{S\sigma\sqrt{T}}, \quad \text{Profit} \approx \frac{1}{2}\Gamma(\Delta S)^2$$

8.7 Module VI — Risk Management (16–18)

8.7.1 Algorithm 16: VaR

$$\text{VaR}_\alpha = z_\alpha \sqrt{w^T \Sigma w}\sqrt{T}$$

8.7.2 Algorithm 17: CVaR

$$\text{CVaR}_\alpha = \mathbb{E}[L \mid L > \text{VaR}_\alpha]$$

8.7.3 Algorithm 18: Maximum Drawdown

$$\text{MDD} = \max_t \frac{P_{\text{peak}}(t) - P_t}{P_{\text{peak}}(t)}$$

8.8 Module VII — Yield Optimization (19–20)

8.8.1 Algorithm 19: Nelson-Siegel Yield Curve

$$y(\tau) = \beta_0 + \beta_1 \frac{1 - e^{-\tau/\lambda}}{\tau/\lambda} + \beta_2 \left(\frac{1 - e^{-\tau/\lambda}}{\tau/\lambda} - e^{-\tau/\lambda} \right)$$

8.8.2 Algorithm 20: Duration and Convexity

$$D_{\text{mod}} = -\frac{1}{P}\frac{dP}{dy}, \quad C = \frac{1}{P}\frac{d^2P}{dy^2}$$

8.9 Module VIII — AMM & Liquidity (21–22)

8.9.1 Algorithm 21: Constant Product AMM

$$k = x \times y, \quad \Delta y = \frac{y \cdot \Delta x}{x + \Delta x}$$

8.9.2 Algorithm 22: Impermanent Loss

$$\text{IL} = \frac{2\sqrt{r}}{1 + r} - 1, \quad r = \frac{P_{\text{new}}}{P_{\text{initial}}}$$

8.10 Module IX — Treasury Management (23–27)

8.10.1 Algorithm 23: Treasury Rebalancing

$$\min_w \sum_i \lambda_i (w_i - w_i^*)^2 + \gamma \sum_i |w_i - w_{i,\text{current}}|$$

8.10.2 Algorithm 24: Multi-Signature Threshold

$$\text{Authorization} = \mathbf{1}\left\{\sum_i \text{sign}_i \geq m\right\}$$

8.10.3 Algorithm 25: Liquidity Buffer

$$\text{Buffer_ratio} = \frac{\text{Liquid_assets}}{\text{Short_term_liabilities}}$$

8.10.4 Algorithm 26: Stablecoin Peg Monitor

$$\text{Deviation} = \frac{P_{\text{market}} - P_{\text{peg}}}{P_{\text{peg}}}$$

8.10.5 Algorithm 27: Collateral Health

$$\text{LTV}_i = \frac{\text{Debt}_i}{\text{Collateral}_i}$$

8.11 Module X — Safeguards (28–31)

8.11.1 Algorithm 28: Circuit Breaker

$$\text{Halt} = \mathbf{1}\left\{\bigvee_j (\text{Metric}_j > \text{threshold}_j)\right\}$$

8.11.2 Algorithm 29: Stress Testing

$$\text{Portfolio_loss}_s = \sum_i w_i \cdot \delta_{s,i} \cdot \text{Exposure}_i$$

8.11.3 Algorithm 30: Fraud Detection

$$\text{Score}(tx) = \sum_j w_j \cdot f_j(tx)$$

8.11.4 Algorithm 31: Audit Trail Generator

$$\text{Record}_t = \{\text{timestamp}, \text{op}, \text{actor}, \text{inputs}, \text{outputs}, H(\text{prev_hash} \| \text{Record}_t)\}$$

8.12 Master Index

Table 8.1: ADR-31 master index.

Alg	Name	Module	Complexity
01	Accrued Interest	I	$O(1)$
02	Make-Whole Premium	I	$O(n)$

Alg	Name	Module	Complexity
03	YTM Solver	I	$O(kn)$
04	Debt-to-Equity Conversion	II	$O(n)$
05	Pro Rata Distribution	II	$O(n)$
06	Fractional Share Liquidation	II	$O(n)$
07	Realized Volatility	III	$O(n)$
08	GARCH(1,1) Forecast	III	$O(n)$
09	Implied Volatility Solver	III	$O(k)$
10	TWAP Execution	IV	$O(N)$
11	VWAP Execution	IV	$O(N)$
12	Implementation Shortfall	IV	$O(1)$
13	Black-Scholes Pricing	V	$O(1)$
14	Dynamic Delta Hedging	V	$O(1)$
15	Gamma Scalping	V	$O(n)$
16	Value-at-Risk	VI	$O(n^2)$
17	Conditional VaR	VI	$O(n \log n)$
18	Maximum Drawdown	VI	$O(n)$
19	Yield Curve Construction	VII	$O(n)$
20	Duration/Convexity	VII	$O(n)$
21	Constant Product AMM	VIII	$O(1)$
22	Impermanent Loss	VIII	$O(1)$
23	Treasury Rebalancing	IX	$O(kn)$
24	Multi-Sig Threshold	IX	$O(m)$
25	Liquidity Buffer Monitor	IX	$O(1)$
26	Stablecoin Peg Monitor	IX	$O(n)$
27	Collateral Health Check	IX	$O(n)$
28	Circuit Breaker	X	$O(1)$
29	Stress Testing	X	$O(kn)$
30	Fraud Detection	X	$O(n \log n)$
31	Audit Trail Generator	X	$O(n)$

Chapter 9

DYREO Capital Allocation

9.1 Mathematical Formulation

$$\max_{\mathbf{w}} \left\{ \alpha \mathbf{w}^T \boldsymbol{\mu} + \beta \mathbf{w}^T \mathbf{s} - \gamma \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} \right\}$$

Subject to:

$$\begin{aligned} \mathbf{w}^T \mathbf{s} &\geq S_{\min} \quad \text{(minimum social impact)} \\ P(\mathbf{w}^T \mathbf{r} < -\delta) &\leq \epsilon \quad \text{(capital preservation)} \\ w_{\text{housing}} &\in [0.40, 0.50] \\ w_{\text{clean_energy}} &\in [0.15, 0.25] \\ w_{\text{fixed_income}} &\in [0.20, 0.30] \\ w_{\text{liquidity}} &\in [0.05, 0.15] \\ \sum_i w_i = 1, &\quad w_i \geq 0 \end{aligned}$$

9.2 Asset Allocation

Table 9.1: DYREO solvent trust asset class structure.

Asset Class	**Allocation**	**Target Yield**
Sustainable Housing	45%	6.5–8.5%
Clean Energy Infra	20%	7.0–9.0%
Sovereign / Fixed Income	25%	4.5–5.5%
Liquid Reserves	10%	4.0–4.5%

9.3 Social Energy Return on Investment

$$\text{S-EROI} = \frac{\int_0^T [P_{\text{solar}}(t) - P_{\text{loss}}(t)] dt}{K_{\text{cap}} + \int_0^T C_{\text{main}}(t) dt} \geq 0.73 \text{ kWh/\$}$$

9.4 Perpetuity Rule

$$D(t) \leq 0.05 \times \frac{1}{5} \sum_{k=1}^{5} \text{Corpus}(t-k)$$

Any excess investment yield above 5% is automatically retained and compounding within the capital preservation core.

9.5 K-S Drift Detection

1. Continuous data ingestion.
2. Two-sample Kolmogorov-Smirnov test:

$$D_n = \sup_x |F_{\text{empirical}}(x) - F_{\text{baseline}}(x)|$$

3. Threshold evaluation: $p < 0.05$?
4. Automated convex re-optimization.

Chapter 10

30-House Solar/BESS Pilot

10.1 Specification

Table 10.1: Scaled 30-house pilot specification.

Component	**Spec**
Houses	30 Baltimore rowhomes
Solar	∼5 kW/house = 150 kW total
BESS	∼10 kWh LFP/house = 300 kWh total
Grid model	Decentralized microgrid; 24-hour islanding
Revenue	Energy service tariffs + affordable housing leases
Municipal target	5,000-unit vacancy reduction by FY2029

10.2 Funding Rail

WY Bitcoin Treasury of USEC funds → regulated OTC/custodian conversion → restricted grant to PSDLEF-S. No leverage, no lending, no derivatives. 3-of-5 multisig with PQC signatures.

10.3 Risk Mitigation

- Contractor compliance: competitive bidding, fixed-cost contracts, performance bonds.
- Grid interconnection: 24-hour islanding capability.
- Tenant stability: affordable municipal rates; revenues swept into trust principal.

Chapter 11

Quantum Gravitational Reactor (QGR) R&D Layer

11.1 Critical Caveat

No commercially operational Quantum Gravitational Reactor exists as of 2026. QGR is integrated as a **high-risk, optionality-preserving R&D layer**, not a load-bearing energy asset.

11.2 Technical Landscape

1. **WARP Reactor (LLNL):** pulsed-power radiation source for MagLIF; real research program.
2. **QGM-NSR:** theoretical neutron superfluid reaction; preprint hypothesis.
3. **G-Engine:** zero-point energy transduction; pseudoscientific by mainstream standards.
4. **LasPina QCDR:** symbolic-curvature reactor; speculative.
5. **Gravitational Wave Emission:** patent EP2233951A1; concept only.

11.3 Pilot Specification

Table 11.1: QGR R&D pilot specification.

Component	**Spec**
Scope	Literature review + 1 bench-scale experiment
Budget	5–10% of total grant
Duration	18–24 months
Deliverable	Peer-reviewed paper OR independent replication
Success gate	Measurable net energy gain
Failure mode	Capital written down; no impact on solar/BESS
Partners	University lab, national lab, or independent replication lab

11.4 Due-Diligence Checklist

1. What is the claimed energy balance (input vs. output)?
2. Has the effect been independently replicated?
3. Is there a peer-reviewed paper describing the mechanism?
4. What is the theoretical basis in established physics?
5. What are the safety and regulatory requirements?

Chapter 12

CUSIP / Maryland Comptroller Transfer

12.1 CUSIP Normalization

The CUSIP check-digit algorithm uses a modified Luhn scheme:

1. Convert characters to numeric values (A=10, …, Z=35).
2. Double values at even positions (0-indexed).
3. Sum the digits of each product.
4. Check digit $= (10 - (\sum \bmod 10)) \bmod 10$.

12.2 Maryland Transfer Protocol

The protocol follows the Maryland Uniform Disposition of Abandoned Property Act (Md. Code Ann., Com. Law §§ 17-101 *et seq.*).

1. Identify all CUSIPs in the ADR portfolio.
2. Validate CUSIP format and check digit.
3. Prepare transfer documentation: authorization letter, proof of ownership, tax ID.
4. Submit via Maryland Unclaimed Property Portal or physical delivery to 80 Calvert Street, Annapolis, MD 21401.
5. Receive State Claim Number.
6. Schedule pickup appointment.

Chapter 13

Dual Court Filing: CIT + COFC

13.1 Strategic Overview

This package consolidates a coordinated dual-court strategy:

- **CIT:** handles trade-administration and transfer aspects under 28 U.S.C. § 1581(i).
- **COFC:** handles pure § 1498 / public-trust declaration.

13.2 Article III Standing

1. **Injury-in-Fact:** deprivation of control and attribution; legal exposure; impairment of open-source distribution.
2. **Causation:** state official interference; federal inaction.
3. **Redressability:** declaratory and injunctive relief would clarify ownership and preempt conflicting state actions.

13.3 Jurisdiction

13.3.1 CIT Residual Jurisdiction

28 U.S.C. § 1581(i): exclusive jurisdiction over civil actions against the United States arising out of federal trade-related laws (EAR, HTSUS, export controls).

13.3.2 COFC § 1498 Jurisdiction

28 U.S.C. § 1498(a): exclusive jurisdiction over claims for compensation for government use of patented inventions.

13.3.3 Transfer Under § 1631

28 U.S.C. § 1631: transfer to cure want of jurisdiction if in the interest of justice.

13.4 Legal Argument

13.4.1 Bayh-Dole Public-Trust Anchoring

35 U.S.C. § 200: federally supported inventions must be utilized to promote free competition and public availability.

13.4.2 Supremacy Clause Preemption

U.S. Const. art. VI, cl. 2: federal law preempts conflicting state law. *United States v. Little Lake Misere Land Co.*, 412 U.S. 580 (1973).

13.4.3 Winter Injunction Standard

Winter v. Natural Resources Defense Council, Inc., 555 U.S. 7 (2008): four-factor test.

13.5 Requested Relief

1. Assume jurisdiction under 28 U.S.C. § 1581(i).
2. Order transfer under 28 U.S.C. § 1631.
3. Declare federal rights under § 1498 and public-trust framework.
4. Enjoin state officials from altering cryptographic logs, asserting conflicting IP claims, or interfering with open-source dissemination.
5. Direct federal agencies to record interests and implement open licensing.
6. Grant leave to proceed in forma pauperis under 28 U.S.C. § 1915.

Chapter 14

Risk Management & Contingency

Table 14.1: Comprehensive risk mitigation framework.

Risk	**Stress Scenario**	**Mitigation**
Macroeconomic	High inflation & rate spikes (>8%)	DYREO shifts to inflation-indexed real assets.
Political	Congressional seizure / diversion	UDTA purpose trust structure; 5th Amendment Takings Clause.
Cybernetic	Quantum decryption / data spoofing	FIPS 203/204/205; OSI-P Merkle chain.
Operational	LSH bucket drift / demographic bias	R8 anomaly detection; fairness jitter ϵ_i.
QGR	Failure to replicate	Capital written down; no impact on solar/BESS.
Legal	Adverse court ruling	Dual-court strategy; state coordination via declaratory relief.

Chapter 15

Implementation Roadmap

Table 15.1: Integrated phased roadmap.

Phase	**Timeline**	**Deliverable**
0	0–3 mo	Trust amendment; WY/USEC funding agreement; counsel opinions
1	3–6 mo	PQR layer live; 1 PQC custody node; QGR literature review
2	6–12 mo	30-house solar/BESS pilot energized; QGR partner selection
3	12–18 mo	MORTAR/OSI-P pilot; 10 filings; genomic pipeline; QGR bench experiment
4	18–24 mo	Audit, KPIs; QGR replication report; scale decision
5	24–36 mo	If QGR validated: pilot integration into 1–2 houses

Chapter 16

Key Performance Indicators

- 30 houses retrofitted; 150 kW PV; 300 kWh BESS.
- 1 PQC custody node; 0 classical public-key primitives.
- 10+ WCAG-compliant filings; 1 Daubert memo.
- 0 breaches; 100% audit-log coverage.
- PQR migration gates passed.
- S-EROI ≥ 0.73 kWh/$.
- QGR: 1 peer-reviewed paper OR 1 independent replication report OR 1 documented failure.
- MORTAR: anti-starvation guarantee $T_{\max}+ \leq 42$ days.
- MORTAR: expedited error rate $\leq$ baseline + 2%.
- LSH: false negative rate $\leq \delta_1$; false positive rate $\leq \delta_2$.

Appendix A

Statutory & Regulatory Reference

- Maryland Uniform Trust Code, Md. Code Ann., Est. & Trusts §§ 14.5-101 *et seq.*
- Md. Code Ann., Est. & Trusts §§ 14.5-402(a)(3)(iii), 14.5-408, 11-102, 15-114, 15-502.
- Md. Code Ann., Com. Law §§ 17-101 *et seq.*, 17-319.
- Md. Rule 15-701; Md. Code Ann., State Gov't § 10-122.
- Uniform Directed Trust Act (UDTA).
- Americans with Disabilities Act, 42 U.S.C. §§ 12101 *et seq.*
- Rehabilitation Act §§ 504, 508, 29 U.S.C. §§ 794, 794d.
- 28 U.S.C. §§ 1491, 1498, 1581(i), 1631, 1915, 2201.
- 35 U.S.C. §§ 200–212 (Bayh-Dole Act).
- 20 C.F.R. § 404.1520a; 20 C.F.R. § 404.1527.
- NIST FIPS 203, 204, 205.
- NSA CNSA 2.0.
- OMB M-23-02, M-24-15.
- ICCPR, 999 U.N.T.S. 171.
- CAT.
- CRPD, G.A. Res. 61/106.
- *Daubert v. Merrell Dow Pharmaceuticals*, 509 U.S. 579 (1993).
- *Olmstead v. L.C.*, 527 U.S. 581 (1999).
- *Winter v. NRDC*, 555 U.S. 7 (2008).
- *Zoltek Corp. v. United States*, 672 F.3d 1309 (Fed. Cir. 2012).
- *United States v. Little Lake Misere Land Co.*, 412 U.S. 580 (1973).
- FRBP 9037; FRE 902(13)–(14); HIPAA; GINA; GDPR.

Appendix B

Algorithm Quick Reference

Table B.1: Algorithm quick reference.

Alg	Name	Purpose
01	Accrued Interest	Bond coupon accrual (30/360, Act/365)
02	Make-Whole Premium	Callable bond redemption premium
03	YTM Solver	Newton-Raphson yield-to-maturity
04	Debt-to-Equity	Restructuring conversion ratio
05	Pro Rata Distribution	Liquidation waterfall allocation
06	Fractional Share	Dust liquidation to cash
07	Realized Volatility	Annualized historical volatility
08	GARCH(1,1)	Conditional volatility forecast
09	Implied Volatility	Black-Scholes inversion
10	TWAP	Time-weighted execution
11	VWAP	Volume-weighted execution
12	Implementation Shortfall	Transaction cost analysis
13	Black-Scholes	European option pricing
14	Delta Hedging	Delta-neutral rebalancing
15	Gamma Scalping	Volatility harvesting
16	VaR	Parametric value-at-risk
17	CVaR	Expected shortfall
18	Maximum Drawdown	Peak-to-trough decline
19	Yield Curve	Nelson-Siegel term structure
20	Duration/Convexity	Interest-rate sensitivity
21	Constant Product AMM	Uniswap v2 invariant
22	Impermanent Loss	LP opportunity cost
23	Treasury Rebalancing	Portfolio weight optimization
24	Multi-Sig Threshold	Governance approval
25	Liquidity Buffer	Basel III LCR analog
26	Stablecoin Peg	Peg deviation monitor
27	Collateral Health	LTV liquidation trigger
28	Circuit Breaker	Trading halt on extreme conditions
29	Stress Testing	Scenario loss simulation

Alg	Name	Purpose
30	Fraud Detection	Anomaly scoring
31	Audit Trail	Hash-chain immutable log

Appendix C

Glossary

ADR Automated Dispute Resolution / Algorithmic Decision Rule.

BESS Battery Energy Storage System.

CBOM Cryptographic Bill of Materials.

CIT Court of International Trade.

COFC Court of Federal Claims.

CRQC Cryptanalytically Relevant Quantum Computer.

CUSIP Committee on Uniform Security Identification Procedures.

DYREO Dynamic Yield & Real Estate Optimization.

eDIB Electronic Disability Claims Processing (SSA).

FES Functional Evidence Score.

GWAS Genome-Wide Association Study.

HSPA-9 Hybrid Semantic Processing Architecture.

LFP Lithium Iron Phosphate.

LSH Locality-Sensitive Hashing.

MORTAR Mortality-Optimized Risk-Triage.

OSI-P OpenSSA Integrity Pipeline.

PQC Post-Quantum Cryptography.

PQR Post-Quantum Resilience & Review.

PRS Polygenic Risk Score.

PSDLEF Public Securities Depository & Ledger Exchange Framework.

QGR Quantum Gravitational Reactor.

SSE-9 Stochastic Semantic Engine.

UDTA Uniform Directed Trust Act.

VPP Virtual Power Plant.

Appendix D

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